Multimodal Techniques for Financial Time-Series Forecasting

Pranav Dhawan Aakash Singh Sivaram Akshit Reddy Palle Sayam Palrecha
Amir Jafari
George Washington University

April 2026

Abstract

Forecasting short-horizon stock movements remains difficult because market prices react to both persistent temporal structure and irregular external events. This paper condenses a larger benchmarking study on S&P 500 equities that compares unimodal, multimodal, and graph-based neural forecasting systems using aligned market prices, news sentiment, and structured fundamentals from SEC filings. The study builds a reduced U.S.-focused version of a large multimodal financial dataset and evaluates recurrent, convolutional, transformer, foundation-model, and graph-attention architectures under a common forecasting task. The benchmark spans classical sequence models, modern token-mixing architectures, pretrained time-series foundation models, and several fusion strategies for integrating textual context. The main finding is that bimodal models combining price trajectories with news sentiment yield the most reliable gains over price-only baselines, while graph models provide useful complementary structure through sector relationships. The results suggest that added complexity is worthwhile when it encodes genuinely complementary information rather than merely increasing parameter count.

1 Introduction

Financial forecasting has long depended on historical price data, yet market behavior is also shaped by news, investor sentiment, and changes in firm fundamentals. Recent multimodal datasets make it possible to study how these heterogeneous signals interact, but they also raise a practical question: which modalities and model families deliver meaningful predictive improvements for short-horizon forecasting?

This paper studies that question for S&P 500 stocks. Given a lookback window of historical prices and optional aligned textual and tabular inputs, the task is to predict the next 5–7 trading-day close prices. The paper presents a broad experimental comparison in which simple temporal baselines, multimodal fusion architectures, and graph neural networks are evaluated within one unified pipeline. The analysis emphasizes the experimental design, the role of cross-modal fusion, and the empirical patterns that matter most for practitioners.

The study is motivated by a central applied concern in financial machine learning: richer data do not automatically translate into better forecasts. Each additional modality introduces its own cadence, noise characteristics, missingness pattern, and alignment burden. As a result, a useful benchmark must evaluate not only whether multimodal learning helps, but also which fusion strategies are robust enough to justify their complexity.

A second motivation is methodological. In many financial applications, predictive pipelines are judged on narrow benchmark gains without sufficiently analyzing why a given model works, under which information conditions it fails, and whether its gains are likely to be operationally meaningful. By broadening the comparison to recurrent networks, patch-based transformers, MLP-style mixers, pretrained time-series foundation models, fusion networks, and graph architectures, the study moves beyond a single-model narrative and instead treats multimodal forecasting as a design space with explicit trade-offs.

Finally, the work is relevant because the target problem is deliberately short horizon. Over a five- to seven-day window, market dynamics are heavily affected by event timing, sentiment shocks, and sector-level co-movement. This setting is therefore well suited to evaluating whether textual and relational context can complement price history in a statistically useful and operationally plausible way.

2 Data and Task Setup

The benchmark is built from the S&P 500 portion of a larger multimodal financial dataset. The full source includes price time series, financial news, corporate filings, and chart images. This paper restricts attention to the U.S. market and excludes image inputs in order to focus on structured and textual signals that can be aligned directly to the forecasting target.

All modalities are mapped to a daily trading calendar at the ticker level. Price-derived variables provide the dense temporal backbone. News is aggregated by publication date and merged into the daily frame, while filing-based fundamental variables are backward-aligned and forward-filled until a newer filing becomes available. After align-

Table 1: Reduced multimodal dataset used in the study.

Type	Size	Records	Samples
Tables (semi-annual)	84.04 GB	2,676	8,028
Time series	1.83 GB	4,213	4,213
News text (daily)	14.1 GB	4,694	3,351,852
Total	108.64 GB	–	3,559,440

ment, synchronized examples are created with fixed look-back windows and short prediction horizons suitable for sequence forecasting.

This alignment step is central to the benchmark's credibility. Financial news is naturally sparse and event-driven, while filing data update only quarterly or semiannually. Without careful temporal alignment, models can either miss useful information or inadvertently exploit information not available at prediction time. The data pipeline addresses this issue by imposing a strict daily index and by carrying forward only historically available filing features. In practical terms, this means each sample reflects the information state that a forecaster could plausibly access on that trading day.

The resulting learning problem can be viewed as conditional sequence forecasting. Let $x_{t-T+1:t}$ denote the look-back window of price-derived inputs, let n_t denote a news representation aligned to the same date range, and let f_t denote the most recently available filing-derived feature vector. The model predicts a short future trajectory $y_{t+1:t+H}$. The question is not merely whether the mapping $(x, n, f) \mapsto y$ can be learned, but whether augmenting x with n and f improves out-of-sample performance enough to justify the additional modeling complexity.

The news modality is particularly useful because articles are mapped to an ordinal sentiment scale that preserves polarity while remaining easy to aggregate at the daily level. In the S&P 500 subset, neutral-to-mildly-positive sentiment constitutes the majority of observations, which helps explain why sentiment-aware models must distinguish between informative event shocks and the much larger mass of routine, low-intensity news.

3 Related Work and Motivation

Three research threads motivate this benchmark. The first is classical financial forecasting, where volatility clustering, autocorrelation structure, and heteroskedastic noise motivate sequence-aware models rather than purely static predictors. Traditional econometric work established that financial time series are noisy, regime-sensitive, and difficult to forecast reliably, which is why even modest and stable improvements are meaningful.

The second thread is modern deep time-series modeling. Architectures such as LSTMs, TCNs, transformers, PatchTST, and TSMixer differ in how they encode temporal dependence, cross-feature interaction, and long-range context. Their inclusion in the benchmark reflects a practical need: before concluding that multimodality helps, one must ensure the unimodal baselines are already competitive.

The third thread is multimodal and relational learning. FinBERT-style encoders enable the extraction of finance-specific sentiment from news text, while graph neural networks provide a way to model dependencies among stocks linked by sector, market conditions, or co-movement structure. Together, these approaches suggest that useful predictive signal may lie not only in a stock's own history, but also in external narrative context and in the behavior of related assets.

4 Modeling Strategy

The benchmark compares three families of approaches.

Unimodal temporal baselines. Price-only models include a stacked LSTM, PatchTST, TSMixer, Chronos foundation models, a TCN–Transformer hybrid, and the Temporal Fusion Transformer (TFT). These architectures test whether stronger temporal inductive biases alone are sufficient for short-horizon forecasting.

Multimodal fusion models. News-enhanced models pair temporal encoders with FinBERT sentiment representations. Three fusion patterns are explored: (i) cross-attention, where temporal states query the news embedding; (ii) late concatenation, where temporal and text embeddings are merged before the prediction head; and (iii) a richer encoder–decoder design combining a CNN price encoder, daily news embeddings, and a transformer decoder.

Relational graph models. Graph neural networks incorporate sector-level or inter-stock dependency structure. Here, GATv2 and EvolveGCN-style ideas are used to propagate information across equities rather than treating each ticker independently.

At a high level, the benchmark progresses from simpler to richer inductive assumptions. The LSTM assumes temporal recurrence and gated memory are sufficient to summarize a stock's recent trajectory. PatchTST instead treats segments of the sequence as token-like objects, allowing attention to operate over coarse temporal units. TSMixer emphasizes alternating mixing across time and feature dimensions, trading some expressive bias for parameter efficiency. Chronos-style models add a foundation-model perspective by transferring priors learned on broad time-series corpora.

The multimodal architectures differ less in raw parameter count than in how they operationalize cross-modal interaction. Cross-attention models allow sentiment to modulate temporal states locally, which is appealing when only a subset of timesteps should react to a textual event. Late-fusion models compress each modality separately before combining them, which is easier to optimize but can sup-

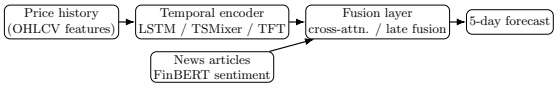

Figure 1: Representative multimodal forecasting pipeline used throughout the paper.

press fine-grained interaction. Encoder–decoder variants preserve sequential structure in both the historical window and the forecast horizon, potentially making them more suitable when the path of the forecast matters as much as the aggregate error.

The comparison highlights an important design distinction between fusion mechanisms. Cross-attention offers asymmetric conditioning: the temporal stream remains primary, while sentiment selectively modulates it. Late fusion is simpler and cheaper, but it risks modality imbalance because the large text embedding can dominate the smaller temporal representation. Hybrid encoder–decoder fusion preserves sequential structure more explicitly, though at a higher modeling and optimization cost.

5 Experimental Protocol

The data pipeline aligns all modalities on a daily trading calendar and constructs synchronized examples at the ticker level. Reported configurations use lookback windows between 30 and 90 trading days and typically predict the next five days, with some variants extending to seven. Price features include engineered OHLCV-based inputs, news is aggregated to daily sentiment-aware representations, and filing variables are carried forward until a newer disclosure arrives.

Evaluation focuses on RMSE, MAE, MAPE, and directional accuracy. This combination is important because financial forecasts can appear numerically accurate while still failing to capture return direction, and conversely may rank movements reasonably well despite calibration error. The paper therefore compares models not only on scale-sensitive regression metrics, but also on whether they correctly anticipate the sign of future movement.

For a set of N forecasted targets with actual values y_i and predictions $\hat{y}_i$, the mean absolute error is

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|,$$

while the mean absolute percentage error is

$$\mathrm{MAPE} = \frac{100}{N}\sum_{i=1}^{N}\left|\frac{y_i - \hat{y}_i}{y_i}\right|.$$

If $r_i = y_i - y_{i-1}$ denotes the realized price change and $\hat{r}_i = \hat{y}_i - y_{i-1}$ the predicted change relative to the latest observed value, then directional accuracy is defined as

$$\mathrm{DirAcc} = \frac{100}{N}\sum_{i=1}^{N}\mathbf{1}[\mathrm{sign}(r_i) = \mathrm{sign}(\hat{r}_i)]\,.$$

These definitions make explicit that a model can be judged both on numerical calibration and on its ability to identify whether the market moved upward or downward over the prediction horizon.

The use of multiple metrics is especially important in finance because error scales can be misleading across modeling spaces. Some models forecast standardized prices, others log returns, and foundation models may report performance under partially different normalization conventions. The benchmark therefore treats metric comparisons with care and uses qualitative cross-model trends rather than over-interpreting any single scalar. This is one reason the paper emphasizes recurring patterns—for example, the repeated value of news sentiment—over isolated leaderboard claims.

Another noteworthy protocol choice is the reliance on residual-style forecasting in some multimodal variants, where the model predicts an increment relative to the last observed price. This can stabilize training by preserving a natural local baseline and asking the model to explain deviation rather than reconstruct the entire future level from scratch. Such a design is particularly sensible when the forecast horizon is short and much of the next-step value is inherited from the current price level.

6 Results

Table 2 summarizes the model comparison. Among price-only baselines, TSMixer and PatchTST outperform the LSTM on RMSE and MAE, indicating that modern token-mixing and patch-based sequence models are strong baselines even without extra modalities. Chronos models also show competitive behavior, though one result is in a different metric scale and should be interpreted cautiously.

The strongest qualitative conclusion is that *price + news* fusion performs best overall. In particular, ChronosFinBert achieves clear gains relative to several unimodal systems, supporting the idea that event-driven textual information adds predictive value beyond historical prices alone. By contrast, structured fundamentals appear more situational: they provide useful context but tend to refresh more slowly and therefore contribute smaller incremental gains in short-horizon settings. Graph models further improve directional reasoning by capturing dependencies among related stocks.

The unimodal comparison is still informative. TSMixer records the strongest RMSE among the directly comparable price-only baselines in the table, while PatchTST also improves substantially over the LSTM. This suggests that architectures designed for long-context temporal representation learning can extract richer structure from market sequences than conventional recurrent encoders, even before any external modality is introduced.

It is also useful to interpret the weaker multimodal rows carefully. Not every price-plus-news architecture outperforms every price-only baseline. This indicates that the

Table 2: Selected forecasting results across the evaluated forecasting models.

Model	Modality	RMSE	MAE	MAPE	Dir. Acc.
LSTM	Price	0.0375	0.0290	206.0	–
PatchTST	Price	0.0236	**0.0144**	106.1	51.4
TSMixer	Price	**0.0223**	0.0147	**19.8**	51.3
Chronos T5-Large*	Price	0.550	0.471	1.89	60.0
GAT	Price+Graph	–	–	–	**∼62**
ChronosFinBert 60d	Price+News	0.125	0.042	171.8	–
TFT	Price	0.0673	0.0350	25.3	–
TCN–Transformer	Price	0.0673	0.0352	25.7	–
TFT–FinBERT	Price+News	1.167	0.750	623.0	–
FinBERT+CNN+Trans.	Price+News	1.45	0.93	–	–
FinBERT–TCN	Price+News	2.25	1.53	–	–

Bold indicates the best value per directly comparable metric column. *Chronos T5-Large is shown on a different metric scale, so its MAPE is not bolded for comparison.

presence of an informative modality is not enough; the architecture must be able to incorporate that modality without destabilizing optimization or overwhelming the temporal representation. In that sense, the benchmark's negative results are just as valuable as its positive ones: they show that multimodality is beneficial only when aligned with an appropriate fusion mechanism.

The graph-based findings point in a related direction. Even when graph models do not dominate every metric, their directional performance suggests that relational structure captures something economically meaningful about synchronous or sector-linked movement. This is particularly relevant in diversified market settings where shocks propagate across related firms rather than remaining isolated to single securities.

7 Discussion

Two broader lessons emerge from the benchmark. First, modality choice matters more than architectural complexity alone. Dense price histories remain essential, but adding high-frequency news sentiment is particularly effective because it captures shocks that are invisible in the trailing price window. Second, the *way* modalities are fused matters. The best-performing systems do not simply append every available feature; instead, they align inputs to the forecasting objective and use fusion mechanisms that preserve the dominance of the temporal signal while still exploiting textual context.

These findings have practical implications. For production-style short-horizon forecasting, a robust bimodal pipeline using prices plus news is likely a better starting point than a heavier multimodal stack that adds sparse or slowly changing variables without careful handling. Graph reasoning is attractive when portfolios or sector-level dependencies matter, but its value depends on whether those relationships are stable and informative over the forecast horizon.

A practical takeaway for deployment is that modality governance matters as much as model selection. News pipelines require timestamp integrity, deduplication, and robust handling of days with no relevant articles. Filing features require explicit treatment of reporting lag. Graph structure requires a clear policy for defining and updating edges. In real systems, these engineering choices often dominate the difference between a benchmark improvement and a usable forecasting tool.

This study has several limitations. First, some models are evaluated in slightly different metric spaces, making direct row-by-row comparison imperfect. Second, structured filing data are inherently lower frequency than daily prices and news, so their contribution may be underestimated in a short-horizon task. Third, the benchmark excludes image inputs, meaning the study evaluates targeted multimodality rather than the full four-modal dataset.

A final limitation is external validity. S&P 500 firms are large, liquid, and information-rich. The relative value of news, fundamentals, and graph structure may look different for small-cap equities, less liquid international assets, or macroeconomic series. Future work could therefore test whether the same ranking of modalities persists across markets, across regime changes, and under decision-oriented objectives such as portfolio sorting or risk-adjusted return.

8 Future Directions

Several promising extensions follow naturally from the benchmark. One is adaptive fusion, where the model learns not only how to combine modalities but also when each modality should be trusted. Such a system could down-weight sparse or noisy news days while amplifying sentiment during earnings releases or major firm-specific events.

A second direction is richer relational modeling. The present graph formulations appear to use relatively coarse sector or cross-sectional structure. Future variants could incorporate dynamic edges derived from rolling correlation, supply-chain relationships, analyst coverage overlap, or co-mention patterns in financial news. This would make the relational component more reflective of how dependencies evolve in live markets.

A third direction is evaluation under action-based criteria. A forecasting model may reduce RMSE while offering little trading or ranking utility, or it may produce modest numerical gains that translate into much better cross-sectional ordering. Extending the benchmark to include ranking loss, top-k selection quality, or portfolio-level backtests would offer a more decision-relevant picture of multimodal value.

A fourth direction is more faithful use of the full four-modal dataset. Because the present study excludes chart images, it deliberately narrows the information space in favor of cleaner temporal alignment. A follow-up study could reintroduce visual signals in a controlled way to test whether technical-chart imagery contains incremental predictive content beyond what is already encoded by nu-

merical price features. That comparison would also help clarify whether image inputs add genuine value or merely duplicate information already available in the time series.

Finally, future work could explore regime-aware training and evaluation. Financial signals are known to vary across calm periods, earnings seasons, crises, and macro-driven dislocations. A model that learns a single global mapping may underperform a system that conditions on regime indicators or adapts its fusion behavior across volatility states. Investigating this question would strengthen the practical relevance of multimodal forecasting systems for live deployment.

9 Broader Implications

Beyond the immediate forecasting task, the study contributes to a broader discussion about how machine learning systems should combine heterogeneous evidence under uncertainty. Financial data are an especially demanding testbed because they are noisy, partially redundant, temporally misaligned, and highly sensitive to regime shifts. A system that performs well in this setting must do more than interpolate within a clean benchmark; it must reconcile information that differs in cadence, semantic content, and reliability.

This is why the benchmark's main message is not simply that more data are better. Instead, it suggests a more disciplined principle: additional modalities help when they contribute information that is both complementary and operationally aligned with the prediction target. In the present case, news sentiment satisfies that criterion more often than slowly refreshed filing variables, while graph structure adds value when cross-asset dependence is central to the task. Framed this way, the paper offers guidance that extends beyond finance to other multimodal sequence problems where information arrives asynchronously and with unequal trustworthiness.

10 Conclusion

This paper presents an empirical study of multimodal stock forecasting on S&P 500 equities. The evidence shows that modern price-only sequence models are strong baselines, but multimodal fusion with financial news delivers the most consistent predictive gains. Structured fundamentals and graph relations remain useful, though their benefits are more context dependent. Overall, the results argue for *targeted multimodality*: additional inputs should be included when they supply complementary information at the same temporal scale as the prediction task.

References

[1] Araci, D. (2019). FinBERT: Financial sentiment analysis with pre-trained language models. *arXiv preprint arXiv:1908.10063*.

[2] Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., Shen, H., & Wang, Y. (2024). Chronos: Learning the language of time series. *arXiv preprint arXiv:2403.07815*.

[3] Barberis, N., & Thaler, R. (2003). A survey of behavioral finance. *Handbook of the Economics of Finance*, 1, 1053–1128.

[4] Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

[5] Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation*, 9(8), 1735–1780.

[6] Nie, Y., Nguyen, N. H., Sinthong, P., & Kalagnanam, J. (2023). A time series is worth 64 words: Long-term forecasting with transformers. *International Conference on Learning Representations (ICLR)*.

[7] Chen, S. A., Li, C. L., Yoder, N., Arik, S. O., & Pfister, T. (2023). TSMixer: An all-MLP architecture for time series forecasting. *Transactions on Machine Learning Research*.

[8] Bai, S., Kolter, J. Z., & Koltun, V. (2018). An empirical evaluation of generic convolutional and recurrent networks for sequence modeling. *arXiv preprint arXiv:1803.01271*.

[9] Alharthi, M. A., Ansari, A. F., Stella, L., Turkmen, C., Zhang, X., Mercado, P., & Wang, Y. (2025). Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2503.12746*.

[10] Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of deep bidirectional transformers for language understanding. *arXiv preprint arXiv:1810.04805*.

[11] Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3), 1139–1168.

[12] Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R., & Le, Q. V. (2019). XLNet: Generalized autoregressive pretraining for language understanding. *Advances in Neural Information Processing Systems (NeurIPS)*.

[13] Baltrušaitis, T., Ahuja, C., & Morency, L. P. (2019). Multimodal machine learning: A survey and taxonomy. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 41(2), 423–443.

[14] Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph attention networks. *International Conference on Learning Representations (ICLR)*.

[15] Brody, S., Alon, U., & Yahav, E. (2022). How attentive are graph attention networks? *International Conference on Learning Representations (ICLR)*.

[16] Pareja, A., Domeniconi, G., Chen, J., Ma, T., Suzumura, A., Kanezashi, H., Kaler, T., & Leiserson, C. E. (2020). EvolveGCN: Evolving graph convolutional networks for dynamic graphs. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(04), 5363–5370.

[17] Fan, Z., Song, Y., Wang, S., Zhang, M., Chen, Z., & Tang, J. (2021). Heterogeneous temporal graph neural network. *Proceedings of the 27th ACM SIGKDD Conference on Knowledge Discovery & Data Mining*, 509–519.

[18] Wenyan, X. (2024). Multimodal dataset for financial time series forecasting: Image, text, table and time series. *Hugging Face Datasets*. Retrieved from `https://huggingface.co/datasets/Wenyan0110/Multimodal-Dataset-Image_Text_Table_TimeSeries-for-Financial-Time-Series-Forecasting`.

[19] Peng, B., Chen, M., & Zhang, J. (2025). FinMultiTime: A four-modal bilingual dataset for financial time-series analysis. *arXiv preprint arXiv:2502.14766*.

[20] Lim, B., Arık, S. O., Loeff, N., & Pfister, T. (2021). Temporal fusion transformers for interpretable multi-horizon time series forecasting. *International Journal of Forecasting*, 37(4), 1748–1764.

[21] Loshchilov, I., & Hutter, F. (2017). Decoupled weight decay regularization. *arXiv preprint arXiv:1711.05101*.

[22] Huber, P. J. (1964). Robust estimation of a location parameter. *Annals of Mathematical Statistics*, 35(1), 73–101.

[23] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*.

[24] Raffel, C., Shazeer, N., Roberts, A., Lee, K., Narang, S., Matena, M., & Liu, P. J. (2020). Exploring the limits of transfer learning with a unified text-to-text transformer. *Journal of Machine Learning Research*, 21(140), 1–67.